

Figure 2: Mean LM losses by context length.

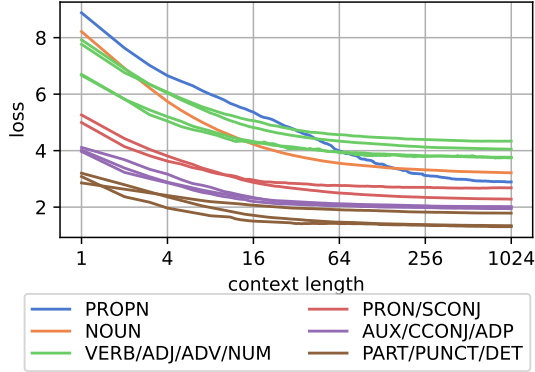


Figure 3: Mean GPT-J loss by context length and part-of-speech (POS) tag of the target token. Only POS tags with at least 100 occurrences in the dataset are included. The tags are grouped (arbitrarily) for clarity.

that very long contexts bring only minimal improvement (though these focused on specific *long-range* architectures and on contexts beyond the range we investigate here).

In Fig. 3, we display the same information (loss by context length) broken down by part-of-speech (POS) tags, for GPT-J only. For most POS tags, the behavior is similar to what we observed in Fig. 2 and the loss appears to stabilize around context lengths 16–64. However, we see a distinct behaviour for proper nouns (PROPN), which are the hardest-to-predict category for short contexts, but whose loss improves steadily with increasing c , surpassing that of regular nouns (NOUN) at $c = 162$ and continuing to improve beyond that point.

4.2 Per-token losses by context length

We have also examined token-level losses, as well as the KL divergence metric (see Section 3.2); an example plot is shown in Fig. 4 and more are found in Appendix C.1. In general, we observe that the values tend to change gradually with c ; large differences are sparse, especially for large c , and can often be attributed to important pieces of information appearing in the context (e.g. “owl” and “swoop” in the context of “birds” in Fig. 4). This justifies

our use of these differences as importance scores.

4.3 Differential importance scores

To facilitate the exploration of Δ -scores from Section 3.3, we have created an interactive web demo,² which allows visualizing the scores for any of the 3 models on the validation set as shown in Fig. 1.

In Fig. 5, we display the magnitudes of the Δ -scores – normalized for each position to sum up to 1 across all context lengths – as a function of context length. The plot suggests a power-law-like inverse relationship where increasing context length proportionally reduces the Δ -score magnitude on average. We interpret this as far-away tokens being less likely to carry information not already covered by shorter contexts. Long contexts (see inset in Fig. 5) bear less importance for larger models than for smaller ones, perhaps because the additional capacity allows relying more on shorter contexts.

In Fig. 6, we also display the mean importance score received by each POS category, by model. We can see that proper nouns (PROPN) are substantially more informative than other categories (which is in line with the observations in the previous section), but less so for the smallest model. This could mean e.g. that larger models are better at memorizing named entities from training data and using them to identify the topic of the document, or simply at copying them from distant context as observed in (Sun et al., 2021).

5 Limitations and future directions

Experiments. We acknowledge the limited scope of our experiments, including only 8 (closed-domain) documents, 3 models and a single language. This is largely due to the limited availability of suitable large LMs and their high computational cost. Still, we believe that our experiments are valuable as a case study that already clearly showcases some interesting features of our methodology.

Computational cost. While we have demonstrated an efficient strategy to obtain predictions for all tokens at all possible context lengths, it still requires running the model N times for a document of length N .

For a k -fold reduction in computational cost, the technique may be modified to use a sliding window with stride $k > 1$ (instead of $k = 1$ as proposed above). See Appendix A.1 for details.

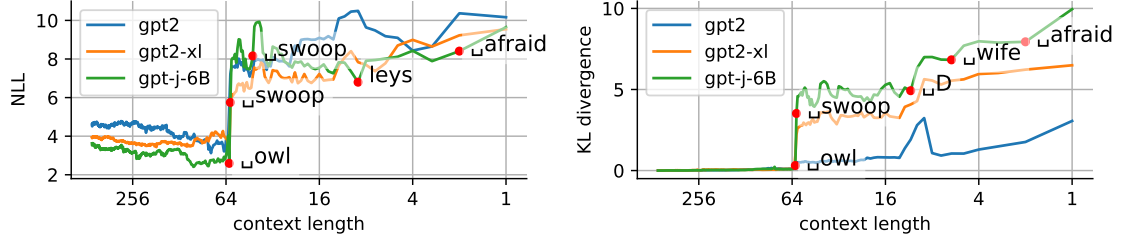


Figure 4: NLL (left) and KL divergence (right) as a function of context length for a selected example: “[...] mortally afraid of **birds**” (same as in Fig. 1). The x axis is reversed for visual correspondence with the left-hand context. The 5 context tokens causing the largest drops in each metric for GPT-J are marked by red dots.

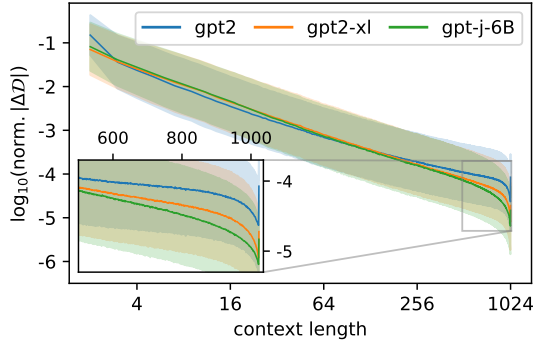


Figure 5: Normalized Δ -score log-magnitude (mean and std. dev.) by context length and by model. Only positions $n \geq 1024$ are included.

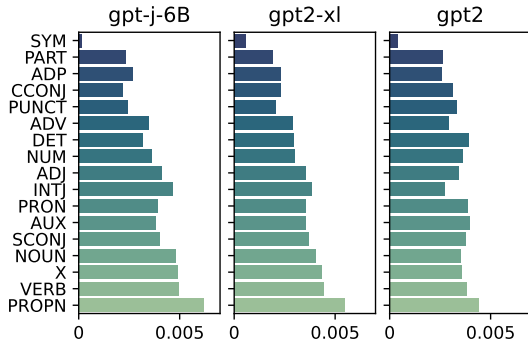


Figure 6: Mean Δ -score by POS tag of the context token and by model.

Choice of metrics. The proposed methodology allows investigating how any given metric is impacted by context, yet our study is limited to NLL loss and the proposed KL divergence metric (the latter for defining importance scores). These may not be optimal for every purpose, and other choices should be explored depending on the application. For example, to study sequences *generated* (sampled) from a LM, one might want to define importance scores using a metric that does depend on the generated token, e.g. its NLL loss or its ranking among all candidates. (Indeed, our web demo also supports Δ -scores defined using NLL loss values.)

6 Conclusion and future directions

We have presented *context length probing*, a novel causal LM explanation technique based on tracking the predictions of the LM as a function of context length, and enabling the assignment of *differential importance scores* (Δ -scores). While it has some advantages over existing techniques, it answers different questions, and should thus be thought of as complementary rather than a substitute.

A particularly interesting feature of our Δ -scores is their apparent potential for discovering *long-range dependencies* (LRDs) (as they are expected to highlight information not already covered by shorter contexts, unlike e.g. attention maps).

Remarkably, our analysis suggests a power-law-like inverse relationship between context length and importance score, seemingly questioning the importance of LRDs in language modeling. While LRDs clearly appear crucial for applications such as long-form text generation, their importance may not be strongly reflected by LM performance metrics like cross entropy or perplexity. We thus believe that there is an opportunity for more specialized benchmarks of LRD modeling capabilities of different models, such as that of Sun et al. (2022), for example. These should further elucidate questions like to what extent improvements in LM performance are due to better LRD modeling, how LRDs are handled by various Transformer variants (e.g. Kitaev et al., 2020; Katharopoulos et al., 2020; Choromanski et al., 2021; Press et al., 2022), or what their importance is for different tasks.

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